Chapter 4. Sampling Distributions and Large-Sample Theory

The Wishart distribution, distributions of $\bar{\mathbf{X}}$ and $\mathbf{S}$, MLE, consistency, multivariate CLT

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1 The Wishart Distribution

Definition 4.1: Wishart distribution

- The Wishart distribution is a family of distributions for symmetric positive definite matrices.
- The distribution of the $p \times p$ random matrix $\mathbf{M} = \mathbf{X}^\top \mathbf{X} = \sum_{i=1}^{n} \mathbf{X}_i \mathbf{X}_i^\top$ is a Wishart distribution.
- With m degrees of freedom,

$$W_p(m \mid \mathbf{\Sigma}) = \text{Wishart with } m \text{ d.f.} = \text{distribution of } \sum_{j=1}^{m} \mathbf{Z}_j \mathbf{Z}_j^\top, \qquad \mathbf{Z}_j \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \mathbf{\Sigma}).$$

We also write $W_p(m, \mathbf{\Sigma})$.

- It is the multivariate extension of χ^2: if $\mathbf{M} \sim W_1(n, \sigma^2)$ then $\mathbf{M}/\sigma^2 \sim \chi^2_n$. E.g. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(0, \sigma^2)$ gives $\mathbf{M} = \sum X_i^2$, $\mathbf{M}/\sigma^2 \sim \chi^2_n$.
- $W_p(n, \mathbf{I})$ is the *standard Wishart* distribution.
- If $\boldsymbol{\mu} = \mathbf{0}$ is known and $\mathbf{S} = \frac{1}{n}\mathbf{X}^\top \mathbf{X} = \frac{1}{n}\sum \mathbf{X}_i \mathbf{X}_i^\top$, then $n\mathbf{S} \sim W_p(n, \mathbf{\Sigma})$.
- If $m \geq p$ and $\mathbf{\Sigma} \succ 0$, $\mathbf{M}$ is PD with probability 1.

Theorem 4.2: properties of the Wishart distribution

(i) For $\mathbf{M} \sim W_p(n, \mathbf{\Sigma})$ and $\mathbf{B}_{(p\times m)}$, $\mathbf{B}^\top \mathbf{M} \mathbf{B} \sim W_m(n, \mathbf{B}^\top \mathbf{\Sigma} \mathbf{B})$.

(ii) For $\mathbf{M} \sim W_p(n, \mathbf{\Sigma})$ with $\mathbf{\Sigma} \succ 0$, $\mathbf{\Sigma}^{-1/2} \mathbf{M} \mathbf{\Sigma}^{-1/2} \sim W_p(n, \mathbf{I}_p)$.

(iii) If $\mathbf{M}_i \sim W_p(n_i, \mathbf{\Sigma})$ $(i = 1, \ldots, k)$ are independent, $\sum_i \mathbf{M}_i \sim W_p(n, \mathbf{\Sigma})$ with $n = n_1 + \cdots + n_k$.

(iv) For $\mathbf{M}_n \sim W_p(n, \mathbf{\Sigma})$, $\mathrm{E}(\mathbf{M}_n) = n\mathbf{\Sigma}$ and $\mathrm{Var}((\mathbf{M}_n)_{ij}) = n(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$. Hence $\hat{\mathbf{\Sigma}}_n = \mathbf{M}_n/n$ has $\mathrm{E}(\hat{\mathbf{\Sigma}}_n) = \mathbf{\Sigma}$ and $\mathrm{Var}[(\hat{\mathbf{\Sigma}}_n)_{ij}] = (\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})/n$.

(v) The law of large numbers and the Cramér–Wold device give $\mathbf{M}_n/n \to \mathbf{\Sigma}$ in probability as $n \to \infty$.

Proof. (i) Write $\mathbf{M} = \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^\top$ with $\mathbf{x}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \mathbf{\Sigma})$. Then $\mathbf{B}^\top \mathbf{M} \mathbf{B} = \sum_i \mathbf{B}^\top \mathbf{x}_i (\mathbf{B}^\top \mathbf{x}_i)^\top$ and $\mathbf{B}^\top \mathbf{x}_i \overset{\text{iid}}{\sim} N_m(\mathbf{0}, \mathbf{B}^\top \mathbf{\Sigma} \mathbf{B})$.

(ii) Special case of (i) with $\mathbf{B} = \mathbf{\Sigma}^{-1/2}$.

(iii) With $\mathbf{M}_i = \sum_j \mathbf{x}_{ij} \mathbf{x}_{ij}^\top$, $\sum_i \mathbf{M}_i = \sum_i \sum_j \mathbf{x}_{ij} \mathbf{x}_{ij}^\top \sim W_p(\sum n_i, \mathbf{\Sigma})$.

(iv) $\mathrm{E}[\mathbf{M}_n] = \mathrm{E}(\sum_k \mathbf{x}_k \mathbf{x}_k^\top) = \sum_k \mathrm{Var}(\mathbf{x}_k) = n\mathbf{\Sigma}$. Terms with different k are independent, so $\mathrm{Var}((\mathbf{M}_n)_{ij}) = n\{\mathrm{E}(X_{ki}^2 X_{kj}^2) - \Sigma_{ij}^2\}$. Using $\mathrm{E}(X_{ki} \mid X_{kj}) = \frac{\Sigma_{ij}}{\Sigma_{jj}} X_{kj}$ and $X_{ki} - \mathrm{E}(X_{ki} \mid X_{kj}) \perp\!\!\!\perp X_{kj}$,

$$\mathrm{E}[(X_{ki}X_{kj})^2] = \mathrm{E}[X_{kj}^2 \mathrm{E}(X_{ki}^2 \mid X_{kj})] = \frac{\Sigma_{ij}^2}{\Sigma_{jj}^2} 3\Sigma_{jj}^2 + \left(\Sigma_{ii} - \frac{\Sigma_{ij}^2}{\Sigma_{jj}}\right)\Sigma_{jj} = 2\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj},$$

so $\mathrm{Var} = n(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$.

(v) By Cramér–Wold it suffices that $\sum_{ij} a_{ij} M_{ij}/n \xrightarrow{p} \sum_{ij} a_{ij} \Sigma_{ij}$ for all a_{ij}. Since $\mathbf{M}_n/n = \frac{1}{n}\sum_k \mathbf{x}_k \mathbf{x}_k^\top$ is an average of iid random matrices, the LLN gives $\sum_{ij} a_{ij} \frac{M_{ij}}{n} \xrightarrow{p} \sum_{ij} a_{ij} \Sigma_{ij}$. ■

2 Wishart Quadratic Forms and the Distribution of S

Theorem 4.3: Wishart quadratic forms

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be iid $N_p(\mathbf{0}, \mathbf{\Sigma})$, $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_n)^\top$, and $\mathbf{A}$ an $n \times n$ real symmetric matrix. Then

$$\mathbf{X}^\top \mathbf{A} \mathbf{X} \sim W_p(r, \mathbf{\Sigma}) \iff \mathbf{A} \text{ is idempotent with } \operatorname{rank}(\mathbf{A}) = r.$$

Partial proof ($\Leftarrow$). Since $\mathbf{A}$ is symmetric idempotent of rank r, $\mathbf{A} = \mathbf{\Gamma}\mathbf{\Lambda}\mathbf{\Gamma}^\top$ with $\mathbf{\Lambda} = \operatorname{diag}(1, \ldots, 1, 0, \ldots, 0)$. Write $\mathbf{A} = \mathbf{P} \begin{pmatrix} \mathbf{I}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \mathbf{P}^\top$ for an $n \times n$ orthogonal $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_n]$, so $\mathbf{A} = \sum_{j=1}^{r} \mathbf{p}_j \mathbf{p}_j^\top$ and

$$\mathbf{X}^\top \mathbf{A} \mathbf{X} = \sum_{j=1}^{r} \mathbf{X}^\top \mathbf{p}_j \mathbf{p}_j^\top \mathbf{X}.$$

Now $\mathbf{X}^\top \mathbf{p}_j = \sum_{i=1}^{n} p_{ij} \mathbf{X}_i \sim N_p(\mathbf{0}, \mathbf{\Sigma})$, since

$$\operatorname{Cov}\Big(\sum_i p_{ij}\mathbf{X}_i\Big) = \mathbf{\Sigma} \cdot \mathbf{p}_j^\top \mathbf{p}_j = \mathbf{\Sigma}, \qquad \operatorname{Cov}\Big(\sum_i p_{ij}\mathbf{X}_i, \sum_i p_{ik}\mathbf{X}_i\Big) = \mathbf{p}_j^\top \mathbf{p}_k \mathbf{\Sigma} = \mathbf{0} \ (j \neq k);$$

being jointly normal, $\mathbf{z}_j = \mathbf{X}^\top \mathbf{p}_j$ are iid $N_p(\mathbf{0}, \mathbf{\Sigma})$. Hence $\mathbf{X}^\top \mathbf{A} \mathbf{X} = \sum_{j=1}^{r} \mathbf{z}_j \mathbf{z}_j^\top \sim W_p(r, \mathbf{\Sigma})$. ($\Rightarrow$ reduces to Theorem 3.20 with $p = 1$: $\mathbf{a}^\top \mathbf{X}^\top \mathbf{A} \mathbf{X} \mathbf{a} \sim (\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}) \chi^2_r$ for every $\mathbf{a}$.) ■

Theorem 4.4: distribution of the sample covariance matrix; $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$

If $\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \mathbf{\Sigma})$, then

$$\bar{\mathbf{X}}_n \sim N_p(\boldsymbol{\mu}, \tfrac{1}{n}\mathbf{\Sigma}), \qquad (n-1)\mathbf{S} = \mathbf{X}^\top \Big(\mathbf{I}_n - \tfrac{1}{n}\mathbf{1}_n \mathbf{1}_n^\top\Big)\mathbf{X} \sim W_p(n-1, \mathbf{\Sigma}), \qquad \bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}.$$

Proof.

Step 1. $\mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$ is symmetric idempotent of rank $n-1$. Since it annihilates $\mathbf{1}_n$, $\mathbf{X}^\top \mathbf{H}_n \mathbf{X} = (\mathbf{X} - \mathbf{1}_n \boldsymbol{\mu}^\top)^\top \mathbf{H}_n (\mathbf{X} - \mathbf{1}_n \boldsymbol{\mu}^\top)$, which reduces to the mean-zero case, and Theorem 4.3 gives $W_p(n-1, \mathbf{\Sigma})$. Independence: choose an orthogonal $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_n]$ with $\mathbf{p}_1 = \mathbf{1}_n/\sqrt{n}$ and let $\mathbf{z}_j = (\mathbf{X} - \mathbf{1}_n \boldsymbol{\mu}^\top)^\top \mathbf{p}_j$. As above $\mathbf{z}_1, \ldots, \mathbf{z}_n \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \mathbf{\Sigma})$, $\mathbf{z}_1 = \sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu})$, and since $\mathbf{H}_n = \sum_{j=2}^{n} \mathbf{p}_j \mathbf{p}_j^\top$, $(n-1)\mathbf{S} = \sum_{j=2}^{n} \mathbf{z}_j \mathbf{z}_j^\top$. The first depends only on $\mathbf{z}_1$, the second only on $\mathbf{z}_2, \ldots, \mathbf{z}_n$. ■

3 A Theorem on Inverse Wishart Ratios

Theorem 4.5: ratios involving an inverse Wishart matrix

Suppose $\mathbf{M} \sim W_p(n, \mathbf{\Sigma})$, $\mathbf{a} \in \mathbb{R}^p$, and $n > p - 1$. Then

(i) $(\mathbf{\Sigma}^{-1})_{11} / (\mathbf{M}^{-1})_{11} \sim \chi^2_{n-p+1}$;

(ii) for any constant $\mathbf{a} \neq \mathbf{0}$, $\dfrac{\mathbf{a}^\top \mathbf{\Sigma}^{-1} \mathbf{a}}{\mathbf{a}^\top \mathbf{M}^{-1} \mathbf{a}} \sim \chi^2_{n-p+1}$;

(iii) if $\mathbf{y} \in \mathbb{R}^p$ is random and independent of $\mathbf{M}$ (with $P(\mathbf{y} = \mathbf{0}) = 0$), then $\dfrac{\mathbf{y}^\top \mathbf{\Sigma}^{-1} \mathbf{y}}{\mathbf{y}^\top \mathbf{M}^{-1} \mathbf{y}} \sim \chi^2_{n-p+1}$ and is independent of $\mathbf{y}$.

Proof. **(i)** Write

$$\mathbf{M} = \begin{pmatrix} m_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \boldsymbol{\sigma}_{12} \\ \boldsymbol{\sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}.$$

By the block-inverse formula,

$$\frac{(\boldsymbol{\Sigma}^{-1})_{11}}{(\mathbf{M}^{-1})_{11}} = \frac{m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21}}{\sigma_{11} - \boldsymbol{\sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}}.$$

Write $\mathbf{M} = \mathbf{X}^\top\mathbf{X} = \sum_{i=1}^{n} \mathbf{x}_i\mathbf{x}_i^\top$ with $\mathbf{x}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2]$ ($\mathbf{x}_1$: first column). Then

$$m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21} = \mathbf{x}_1^\top\mathbf{x}_1 - \mathbf{x}_1^\top\mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{x}_1 = \mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1,$$

$$\mathbf{A} = \mathbf{A}(\mathbf{X}_2) = \mathbf{I}_n - \mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top.$$

Given $\mathbf{X}_2$, $\mathbf{x}_1 \mid \mathbf{X}_2 \sim N_n(\mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21},\ \sigma_{11\cdot 2}\mathbf{I}_n)$, $\sigma_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}$. Since $\mathbf{A}\mathbf{X}_2 = \mathbf{0}$, $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 = (\mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21})^\top\mathbf{A}(\mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21})$. For any given $\mathbf{X}_2$, $\mathbf{A}$ is symmetric idempotent with $\operatorname{rank}(\mathbf{A}) = \operatorname{tr}(\mathbf{A}) = n - \operatorname{tr}\{(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{X}_2\} = n - (p-1)$, so Theorem 3.20 gives

$$\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 \mid \mathbf{X}_2 \sim \sigma_{11\cdot 2}\,\chi^2_{n-p+1}.$$

This does not depend on $\mathbf{X}_2$, so marginally $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 \sim \sigma_{11\cdot 2}\chi^2_{n-p+1}$, independent of $\mathbf{X}_2$. Hence the ratio is $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1/\sigma_{11\cdot 2} \sim \chi^2_{n-p+1}$.

(i)⇒(ii) Given $\mathbf{a}$, take a $p \times p$ orthogonal $\mathbf{P}$ whose first column is $\mathbf{a}/\|\mathbf{a}\|$, i.e. $\mathbf{a}^\top\mathbf{P} = \mathbf{e}_1^\top$ (w.l.o.g. $\|\mathbf{a}\| = 1$). Then

$$\frac{\mathbf{a}^\top\boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top\mathbf{M}^{-1}\mathbf{a}} = \frac{\mathbf{a}^\top\mathbf{P}\mathbf{P}^\top\boldsymbol{\Sigma}^{-1}\mathbf{P}\mathbf{P}^\top\mathbf{a}}{\mathbf{a}^\top\mathbf{P}\mathbf{P}^\top\mathbf{M}^{-1}\mathbf{P}\mathbf{P}^\top\mathbf{a}} = \frac{\mathbf{e}_1^\top\boldsymbol{\Sigma}_{\mathbf{P}}^{-1}\mathbf{e}_1}{\mathbf{e}_1^\top\mathbf{M}_{\mathbf{P}}^{-1}\mathbf{e}_1}, \quad \boldsymbol{\Sigma}_{\mathbf{P}}^{-1} = \mathbf{P}^\top\boldsymbol{\Sigma}^{-1}\mathbf{P},\ \mathbf{M}_{\mathbf{P}}^{-1} = \mathbf{P}^\top\mathbf{M}^{-1}\mathbf{P}.$$

By Theorem 4.2, $\mathbf{M}_{\mathbf{P}} = \mathbf{P}^\top\mathbf{M}\mathbf{P} \sim W_p(n, \boldsymbol{\Sigma}_{\mathbf{P}})$, so (i) applies: $(\boldsymbol{\Sigma}_{\mathbf{P}}^{-1})_{11}/(\mathbf{M}_{\mathbf{P}}^{-1})_{11} \sim \chi^2_{n-p+1}$.

(ii)⇒(iii) Conditioning on $\mathbf{y} = \mathbf{a}$ leaves the law of $\mathbf{M}$ unchanged (independence), so by (ii) the conditional distribution is χ^2_{n-p+1} for almost all $\mathbf{a} \neq \mathbf{0}$. It does not depend on $\mathbf{a}$, so it is also the marginal law and is independent of $\mathbf{y}$. ■

Remark 4.6: residuals and projections in multivariate analysis

1. The residual of $\mathbf{x}_1$ after projecting onto the span of $\mathbf{X}_2$ is $\mathbf{x}_1^* = \mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \mathbf{x}_1 - \mathrm{BLP}(\mathbf{x}_1 \mid \mathbf{X}_2)$.
2. $\mathbf{x}_1^*$ is orthogonal to (uncorrelated with) $\mathbf{X}_2$; being Gaussian, they are independent.
3. The partial variance $\sigma_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{21}^\top\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \operatorname{Var}(\mathbf{x}_1^*)$ is the variance left after removing $\mathbf{X}_2$.

The residual sum of squares has $n - (p-1)$ degrees of freedom. This theorem is the key to the F distribution of Hotelling's T^2 (Ch. 5).

4 Maximum Likelihood Estimation

Theorem 4.7: MLE of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be iid $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $n > p$. The MLEs are

$$\hat{\boldsymbol{\mu}}_{\text{MLE}} = \bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i, \qquad \hat{\boldsymbol{\Sigma}}_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top =: \mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}.$$

The MLE jointly minimizes the negative log-likelihood

$$\begin{aligned}\ell(\boldsymbol{\mu}, \boldsymbol{\Sigma} \mid \mathbf{x}^n) := -\log L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) &= \frac{n}{2}\log|\boldsymbol{\Sigma}| + \frac{1}{2}\sum_{i=1}^{n}(\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) + c \\ &= \frac{n}{2}\log|\boldsymbol{\Sigma}| + \frac{n}{2}(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) + \frac{1}{2}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) + c.\end{aligned}$$

(Expand $\mathbf{x}_i - \boldsymbol{\mu} = (\mathbf{x}_i - \bar{\mathbf{x}}) + (\bar{\mathbf{x}} - \boldsymbol{\mu})$; the cross term vanishes.)

Step 1 (profile out $\boldsymbol{\mu}$). ℓ depends on $\boldsymbol{\mu}$ only through $(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \geq 0$, minimized by $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ for every $\boldsymbol{\Sigma}$. With $\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) = \operatorname{tr}\{\boldsymbol{\Sigma}^{-1}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top\} = n\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$,

$$2\ell(\hat{\boldsymbol{\mu}}_{\text{MLE}}, \boldsymbol{\Sigma}) = n\log|\boldsymbol{\Sigma}| + n\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) + c,$$

so it remains to minimize $f(\boldsymbol{\Sigma}) = \log|\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$ over $\boldsymbol{\Sigma} \succ 0$.

First proof: von Neumann's inequality. Decompose $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ and $\mathbf{S}_0 = \mathbf{V}\mathbf{D}\mathbf{V}^\top$ (λ_i, d_i in decreasing order; $d_i > 0$ a.s. when $n > p$). The eigenvalues of $\boldsymbol{\Sigma}^{-1}$ in increasing order are $1/\lambda_1 \leq \cdots \leq 1/\lambda_p$, so von Neumann's lower bound gives

$$\log|\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) \geq \log(\lambda_1 \cdots \lambda_p) + \sum_{i=1}^{p}\frac{d_i}{\lambda_i} = \sum_{i=1}^{p}\Big(\log\lambda_i + \frac{d_i}{\lambda_i}\Big),$$

with equality iff $\mathbf{U} = \mathbf{V}$. Let $a_i = d_i/\lambda_i$. Then

$$\sum_{i=1}^{p}\Big(\log\lambda_i + \frac{d_i}{\lambda_i}\Big) = \sum_{i=1}^{p}(\log d_i + a_i - \log a_i) = \log|\mathbf{S}_0| + \sum_{i=1}^{p}(a_i - \log a_i) \geq \log|\mathbf{S}_0| + p,$$

since $x - \log x \geq 1$. The bound is attained iff (1) $\mathbf{U} = \mathbf{V}$ and (2) $a_i = 1$ for all i, i.e. $\boldsymbol{\Lambda} = \mathbf{D}$: $\hat{\boldsymbol{\Sigma}}_{\text{MLE}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top = \mathbf{V}\mathbf{D}\mathbf{V}^\top = \mathbf{S}_0$. ■

Second proof: matrix calculus. Reparametrise by the precision matrix $\mathbf{K} = \boldsymbol{\Sigma}^{-1}$ ($\boldsymbol{\Sigma} \mapsto \boldsymbol{\Sigma}^{-1}$ is a bijection of the PD cone, so by the change-of-variables part of Theorem 2.22 the optimizers correspond). $g(\mathbf{K}) = -\log|\mathbf{K}| + \operatorname{tr}(\mathbf{K}\mathbf{S}_0)$ is defined on the open convex set $\{\mathbf{K} \succ 0\}$, and for symmetric $\mathbf{H} \neq \mathbf{0}$

$$D^2 g(\mathbf{K})[\mathbf{H}] = \operatorname{tr}(\mathbf{K}^{-1}\mathbf{H}\mathbf{K}^{-1}\mathbf{H}) = \|\mathbf{K}^{-1/2}\mathbf{H}\mathbf{K}^{-1/2}\|_F^2 > 0,$$

so g is strictly convex (Theorem 2.21). The first-order condition

$$Dg(\mathbf{K})[\mathbf{H}] = -\operatorname{tr}(\mathbf{K}^{-1}\mathbf{H}) + \operatorname{tr}(\mathbf{S}_0\mathbf{H}) = \operatorname{tr}\{(\mathbf{S}_0 - \mathbf{K}^{-1})\mathbf{H}\} = 0 \quad \forall \mathbf{H}$$

and Theorem 2.20 give $\mathbf{K}^{-1} = \mathbf{S}_0$, the unique minimizer. By profiling (Theorem 2.22), $(\bar{\mathbf{x}}, \mathbf{S}_0)$ is the joint minimizer. ■

Remark 4.8: invariance and the maximized likelihood

A function of the MLE is the MLE of the function: e.g. the eigenvalues of $\mathbf{S}_0$ estimate those of $\boldsymbol{\Sigma}$, and r_{ij} is the MLE of ρ_{ij}. The maximum is $L(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}) = (2\pi)^{-np/2}|\mathbf{S}_0|^{-n/2}e^{-np/2}$, which depends on the data only through the generalized variance $|\mathbf{S}_0|$; this is used for likelihood ratio tests (Ch. 5).

5 Large-Sample Behavior of $\bar{\mathbf{X}}$ and $\mathbf{S}$

Theorem 4.9: law of large numbers

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n \in \mathbb{R}^p$ be iid with $\mathrm{E}(\mathbf{X}_j) = \boldsymbol{\mu}$ and finite $\mathrm{Cov}(\mathbf{X}_j) = \boldsymbol{\Sigma}$. Then, as $n \to \infty$,

$$\text{(a) } \bar{\mathbf{X}}_n \xrightarrow{p} \boldsymbol{\mu}, \qquad \text{(b) } \mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}, \qquad \text{(c) } \mathbf{S}_0 \xrightarrow{p} \boldsymbol{\Sigma}.$$

Moreover $\bar{\mathbf{X}}_n$ and $\mathbf{S}$ are unbiased.

Proof. **(a)** By Markov's inequality,

$$P(\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\| > \epsilon) \le \frac{1}{\epsilon^2}\mathrm{E}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|^2 = \frac{1}{\epsilon^2}\mathrm{E}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) = \frac{1}{\epsilon^2}\mathrm{E}[\mathrm{tr}\{(\bar{\mathbf{X}}_n - \boldsymbol{\mu})(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top\}]$$
$$= \frac{1}{\epsilon^2}\mathrm{tr}(\mathrm{Var}(\bar{\mathbf{X}}_n)) = \frac{\mathrm{tr}(\boldsymbol{\Sigma})}{n\epsilon^2} \xrightarrow[n\to\infty]{} 0.$$

(b) Adding and subtracting $\boldsymbol{\mu}$ inside each factor and using $\sum_i(\mathbf{X}_i - \boldsymbol{\mu}) = n(\bar{\mathbf{X}} - \boldsymbol{\mu})$,

$$(n-1)\mathbf{S} = \sum_{i=1}^n(\mathbf{X}_i - \bar{\mathbf{X}}_n)(\mathbf{X}_i - \bar{\mathbf{X}}_n)^\top = \sum_{i=1}^n(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top.$$

Hence

$$\frac{n-1}{n}\mathbf{S} = \frac{1}{n}\sum_{i=1}^n(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - (\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \xrightarrow{p} \boldsymbol{\Sigma} - \mathbf{0} = \boldsymbol{\Sigma}.$$

The first term is an average of iid matrices with mean $\mathrm{E}(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top = \boldsymbol{\Sigma}$, so it converges to $\boldsymbol{\Sigma}$ by the weak LLN applied entrywise (matrix convergence is entrywise, Theorem 2.24). The second term tends to $\mathbf{0}$ by (a) and the CMT ($\mathbf{x} \mapsto \mathbf{x}\mathbf{x}^\top$ is continuous). Since $n/(n-1) \to 1$, $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$.

(c) $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$, which was shown to converge to $\boldsymbol{\Sigma}$ in (b).

Unbiasedness. $\mathrm{E}\bar{\mathbf{X}}_n = \boldsymbol{\mu}$, and taking expectations in the identity of (b), $\mathrm{E}\{(n-1)\mathbf{S}\} = n\boldsymbol{\Sigma} - n\mathrm{Var}(\bar{\mathbf{X}}_n) = (n-1)\boldsymbol{\Sigma}$. Under normality, $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$ and Theorem 4.2(iv) also gives the rate $\mathrm{E}\|\mathbf{S} - \boldsymbol{\Sigma}\|_F^2 = \sum_{ij}(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})/(n-1) \to 0$. ■

Remark 4.10: rate of the estimation error

$\mathrm{E}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2^2 = \mathrm{tr}\boldsymbol{\Sigma}/n$, so $E_{\bar{u}} = \|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2$ decays at rate $n^{-1/2}$; so does $\|\mathbf{S} - \boldsymbol{\Sigma}\|_F$ under normality. On a log–log plot of $E_{\bar{u}}, E_S, E_R$ against n the slope is about $-1/2$.

Theorem 4.11: multivariate central limit theorem

Let $\mathbf{X}_1, \mathbf{X}_2, \cdots \in \mathbb{R}^p$ be iid with $\mathrm{E}(\mathbf{X}_i) = \boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{X}_i) = \boldsymbol{\Sigma}$. Then

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n(\mathbf{X}_i - \boldsymbol{\mu}) \Rightarrow N_p(\mathbf{0}, \boldsymbol{\Sigma}), \qquad \text{equivalently} \qquad \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \Rightarrow N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

Proof. By Cramér–Wold it suffices to show $\mathbf{t}^\top\sqrt{n}(\bar{\mathbf{X}}_n-\boldsymbol{\mu})\Rightarrow\mathbf{t}^\top\mathbf{Y}$, $\mathbf{Y}\sim N_p(\mathbf{0},\boldsymbol{\Sigma})$, for every $\mathbf{t}$. $W_i=\mathbf{t}^\top(\mathbf{X}_i-\boldsymbol{\mu})$ are iid scalars with mean 0 and variance $\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t}$, so the univariate CLT gives $\sqrt{n}\bar{W}_n\Rightarrow N(0,\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$, the law of $\mathbf{t}^\top\mathbf{Y}$ (if $\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t}=0$, $W_i=0$ a.s. and the claim is trivial). ■

Theorem 4.12: large-sample chi-square

Under the same assumptions with $\boldsymbol{\Sigma}\succ 0$, $T_n:=n(\bar{\mathbf{X}}_n-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}}_n-\boldsymbol{\mu})\Rightarrow\chi^2_p$. (Where $\mathbf{S}$ is singular, read $\mathbf{S}^{-1}$ as $r(\mathbf{S})$ defined in the proof.)

Proof. Let $\mathbf{Z}_n:=\sqrt{n}(\bar{\mathbf{X}}_n-\boldsymbol{\mu})$ and $\mathbf{Z}\sim N_p(\mathbf{0},\boldsymbol{\Sigma})$.

(i) $\mathbf{Z}_n\Rightarrow\mathbf{Z}$. For every $\mathbf{t}\in\mathbb{R}^p$, $\mathbf{t}^\top\mathbf{Z}_n=\sqrt{n}(\mathbf{t}^\top\bar{\mathbf{X}}_n-\mathbf{t}^\top\boldsymbol{\mu})\Rightarrow N(0,\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$ by the univariate CLT, so $\mathbf{Z}_n\Rightarrow\mathbf{Z}$ by Cramér–Wold (this is Theorem 4.11).

(ii) Handling a possibly singular S. Let $A_n:=\{\mathbf{S}\text{ is invertible}\}$. Since $\mathbf{S}\xrightarrow{p}\boldsymbol{\Sigma}$ (Theorem 4.9(b)) and $\mathbf{A}\mapsto|\mathbf{A}|$ is continuous, $|\mathbf{S}|\xrightarrow{p}|\boldsymbol{\Sigma}|>0$, hence

$$P(A_n)\geq P(|\mathbf{S}|\geq\tfrac{1}{2}|\boldsymbol{\Sigma}|)\longrightarrow 1.$$

Define $r(\mathbf{A})=\mathbf{A}^{-1}$ if $|\mathbf{A}|\neq 0$ and $r(\mathbf{A})=\mathbf{I}_p$ otherwise. By the definition of r,

$$T_n=n(\bar{\mathbf{X}}-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}}-\boldsymbol{\mu})\,I_{A_n}+n(\bar{\mathbf{X}}-\boldsymbol{\mu})^\top(\bar{\mathbf{X}}-\boldsymbol{\mu})\,I_{A_n^c}=\mathbf{Z}_n^\top r(\mathbf{S})\mathbf{Z}_n,$$

and T_n coincides with the statistic of the theorem on A_n, an event of probability tending to one.

(iii) $r(\mathbf{S})\xrightarrow{p}\boldsymbol{\Sigma}^{-1}$. Since $\mathbf{S}\xrightarrow{p}\boldsymbol{\Sigma}$, by the CMT it suffices that r is continuous at the point $\boldsymbol{\Sigma}$. Since $|\boldsymbol{\Sigma}|>0$ and $\{\mathbf{A}:|\mathbf{A}|\neq 0\}$ is open, $r(\mathbf{A})=\mathbf{A}^{-1}=\operatorname{adj}(\mathbf{A})/|\mathbf{A}|$ on a neighborhood of $\boldsymbol{\Sigma}$ (Section 1 of Ch. 2), whose entries are polynomials divided by a nonvanishing polynomial; so r is continuous at $\boldsymbol{\Sigma}$ and $r(\mathbf{S})\xrightarrow{p}\boldsymbol{\Sigma}^{-1}$.

(iv) Conclusion. Let $Q(\mathbf{z},\mathbf{A})=\mathbf{z}^\top\mathbf{A}\mathbf{z}$, so $T_n=Q(\mathbf{Z}_n,r(\mathbf{S}))$. By Slutsky, $(\mathbf{Z}_n,r(\mathbf{S}))\Rightarrow(\mathbf{Z},\boldsymbol{\Sigma}^{-1})$, and Q is continuous, so by the CMT

$$T_n=Q(\mathbf{Z}_n,r(\mathbf{S}))\Rightarrow Q(\mathbf{Z},\boldsymbol{\Sigma}^{-1})=\mathbf{Z}^\top\boldsymbol{\Sigma}^{-1}\mathbf{Z}\sim\chi^2_p$$

(Example 3.22). ■

This is the basis of large-sample tests and confidence regions for the mean without normality (Theorems 5.16 and 5.17).

Theorem 4.13: asymptotic normality of $h(\bar{\mathbf{X}}_n)$ (delta method)

For $h:\mathbb{R}^p\to\mathbb{R}$ differentiable at $\boldsymbol{\mu}$ with gradient $\nabla h(\boldsymbol{\mu})$,

$$\sqrt{n}\{h(\bar{\mathbf{X}}_n)-h(\boldsymbol{\mu})\}\Rightarrow N_p\big(0,\ (\nabla h(\boldsymbol{\mu}))^\top\boldsymbol{\Sigma}(\nabla h(\boldsymbol{\mu}))\big).$$

Proof. Taylor: $h(\bar{\mathbf{X}}_n)=h(\boldsymbol{\mu})+(\nabla h(\boldsymbol{\mu}))^\top(\bar{\mathbf{X}}_n-\boldsymbol{\mu})+O_p(\|\bar{\mathbf{X}}_n-\boldsymbol{\mu}\|_2^2)$. Multiply by $\sqrt{n}$: $\sqrt{n}(h(\bar{\mathbf{X}}_n)-h(\boldsymbol{\mu}))=(\nabla h(\boldsymbol{\mu}))^\top\sqrt{n}(\bar{\mathbf{X}}_n-\boldsymbol{\mu})+O_p(\sqrt{n}\|\bar{\mathbf{X}}_n-\boldsymbol{\mu}\|^2)$, and the remainder is $o_p(1)$. Slutsky gives $\Rightarrow(\nabla h(\boldsymbol{\mu}))^\top N_p(\mathbf{0},\boldsymbol{\Sigma})$. ■

6 Summary: Distributional Facts Used Later

Fact	Source
$\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$	Theorem 3.7
$(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$, $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$	Theorem 4.3
$(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_p$	Theorem 3.21, Example 3.22
$\mathbf{a}^\top \boldsymbol{\Sigma}^{-1} \mathbf{a} / \mathbf{a}^\top \mathbf{M}^{-1} \mathbf{a} \sim \chi^2_{m-p+1}$, also for independent random $\mathbf{y}$	Theorem 4.5
$n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}) \Rightarrow \chi^2_p$ (no normality)	Theorem 4.11 + Slutsky

Further topics: assessing normality (Q–Q plots; the χ^2 plot of $d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$ against χ^2_p quantiles), detecting outliers (large d_i^2), transformations to near normality (Box–Cox).